

Platform Architecture for GE-LAV Computation

Session 18 • How an LP analyst would build the system

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

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What we'll cover today

1

System requirements

What we need to compute and serve

2

Data layer

Sources, frequency, validation

3

Modeling layer

$L(t)$ calibration, $\pi(L,T)$ computation

4

Output layer

Reports, dashboards, alerts

5

Build vs. buy decision

Open source vs. proprietary

End-to-end platform architecture (conceptual)

LAYER 5: OUTPUT

IC memos • Allocation dashboards • Stress scenarios • Regulatory reports • Risk alerts

LAYER 4: ANALYTICS

Asset valuation • Optimal exit boundary • Portfolio allocation • Risk attribution

LAYER 3: MODELING

OU calibration • $\pi(L,T)$ surface • Jensen correction • McKean-Vlasov solver

LAYER 2: PROCESSING

Liquidity state estimation • Term-structure fitting • Validation tests

LAYER 1: DATA

Lazard • Evercore • Greenhill • Preqin • Pitchbook • Bloomberg • Internal LP data

Build vs. buy: realistic decision matrix

GE-LAV platform building is not a small project. Realistic options:

Pure build (in-house) <i>Best for: large institutional investors (\$10B+ AUM)</i>	12-18 months • 4-6 quant developers • \$2-4M annual cost	<i>Maximum customization, full IP ownership</i>
Buy + customize (vendor) <i>Best for: mid-sized institutional (\$1-10B AUM)</i>	3-6 months • 1-2 internal • \$200-500K annual license	<i>Faster, standardized methodology, vendor lock-in</i>
Open-source stack <i>Best for: research institutions, smaller AUMs</i>	Variable • 1 quant • ~\$50K annual infra cost	<i>Maximum transparency, requires internal expertise</i>
Outsource (consultants) <i>Best for: occasional ad-hoc analyses</i>	Per-project • \$50-500K each	<i>Lowest commitment, doesn't build internal capability</i>

Session 18 summary

What we accomplished today

- 1 GE-LAV platform requires 5 architectural layers: data, processing, modeling, analytics, output
- 2 Calibration of $L(t)$ is data-intensive — needs Lazard/Evercore + internal LP data
- 3 Realistic options: pure build (large), buy+customize (mid), open source (small), outsource (occasional)
- 4 Recommended for course project: open-source implementation in Python with Quantopian-style notebook

Next session

Unit 5 begins: mathematical intuition bridges. Session 19: Brownian motion and Itô calculus